BITCOIN (BTC-USD) QUANTITATIVE STRATEGY & BACKTESTING MASTER REPORT

Comprehensive Multi-Timeframe Confluence, 120-Strategy Universe, Cross-Resolution Benchmark & Complete How-to-Backtest Guide (2014 – 2026)

Asset: Bitcoin (BTC-USD)	**Horizon: 2014-09 to 2026-09 (4,370 Bars)**	**Initial Capital: $10,000 USD**	**Cash Yield: 4.0% Risk-Free Rate**	**Rigor: Zero-Lookahead (T-1 / W-1)**

EXECUTIVE SUMMARY & KEY RESEARCH FINDINGS:

- **The Single-Timeframe Champion:** *Daily Heikin-Ashi Green + 50 SMA* achieved a **2.00 Sharpe Ratio**, generating **83.16% CAGR** with a **-41.55% Max Drawdown** (vs. Buy & Hold's -83.40%), turning $10k into **$7,822,747** while in the market only 39.6% of the time.
- **The Highest Wealth Generator:** *Daily 40 SMA* generated **96.06% CAGR**, compounding $10k into **$16,550,330** (outperforming Buy & Hold by +$13.0M) with a 1.94 Sharpe Ratio.
- **The Multi-Timeframe (MTF) Confluence Champion:** *Weekly 20 SMA Band + Daily 40 SMA* produced **82.52% CAGR**, a **1.83 Sharpe**, a **2.07 Calmar Ratio**, and contained maximum drawdown to **-39.77%** with only 41.8% market exposure.
- **Ultra-Low Drawdown Treasury Shield:** Alexander Elder's *Triple Screen 5* (Weekly MACD Slope + Daily RSI<40 + Daily Donchian) suppressed maximum drawdown to just **-17.79%** over 11 years (CAGR 26.54%, Time in Market 11.5%).
- **Cross-Resolution Degradation Law:** Standalone higher-timeframe execution suffers monotonic degradation due to exit lag (e.g. Daily MACD Sharpe 1.48 decays to 1.25 on 3D and 0.97 on 1W). Weekly signals must serve as *macro permission shields*, never standalone triggers.

1. Bitcoin Macro Market Structure & Cycle Dynamics (2014 – 2026)

Bitcoin's price history is governed by programmatic supply-side 4-year halving cycles (2016, 2020, 2024), intertwined with global liquidity waves and institutional adoption phases. While delivering an extraordinary 70.26% annualized buy-and-hold return, Bitcoin exposes passive investors to catastrophic peak-to-trough drawdowns (-84% in 2018, -77% in 2022, -62% in March 2020). Active quantitative trend-following creates asymmetric capital protection by participating in parabolic expansions while rotating into risk-free cash during structural bear markets.

Figure 1: Bitcoin (BTC-USD) 11-Year Macro Price Action, Halving Cycles, 200 SMA Shield, and Historical Drawdown Phases (2014-2026).

2. The Daily 120-Strategy Quantitative Universe

To rigorously map the parameter space, 120 distinct quantitative strategies were evaluated across six classical and modern systematic families on continuous daily data (September 2015 – September 2026, 4,021 bars):

- **1. Moving Average Trend Following (34 Systems):** Single SMA/EMA filters (periods 20 to 350), Dual Moving Average crossovers, Triple Ribbon systems, and 20W/21W Bull Market Support Bands.
- **2. Momentum & Relative Strength (25 Systems):** RSI mean-reversion, RSI 60/40 and 50/40 bull/bear regimes, Trend+RSI confluences, Stochastic crossovers, and Rate of Change (ROC) filters.
- **3. Volatility Channels & Breakouts (22 Systems):** Donchian Channel breakouts (20H/10L, 50H/20L, 55H/20L Turtle System 2, 100H/30L), Bollinger Bands breakouts and squeezes, Keltner Channels (1.5x, 2.0x, 2.5x ATR), and Chandelier ATR exits.
- **4. MACD & Trend Oscillators (15 Systems):** Standard MACD (12/26/9), Fast MACD (8/17/9), Slow MACD (24/52/18), Zero-line crosses, and multi-indicator confluence filters.
- **5. Trailing Stop Loss & Risk Management (14 Systems):** 200 SMA and Golden Cross systems paired with fixed 10% to 30% Trailing Stop Losses (TSL) and Parabolic SAR.
- **6. Heikin-Ashi & Hybrid Systems (10 Systems):** Pure Heikin-Ashi sequences, HA + SMA/EMA filters, HA + RSI regimes, and quality momentum / volatility ratios.

Top 15 Daily Strategies by Sharpe Ratio (Zero-Lookahead, $10k Initial Capital)

Rank	Strategy Name	Family	CAGR	Max DD	Sharpe	Calmar	Final Value ($)	Exposure
1	Heikin-Ashi Green + 50 SMA Trend	Heikin-Ashi / Hybrid	**83.2%**	-41.5%	**2.00**	2.00	$7,822,747	39.6%
2	SMA Trend (40)	Moving Average	**96.1%**	-62.9%	**1.94**	1.53	$16,550,330	56.7%
3	Fast MACD + 100 SMA Filter	MACD / Oscillators	**73.5%**	-38.2%	**1.84**	1.92	$4,298,259	30.6%
4	ROC(20) > 0 with 200 SMA	Momentum / RSI	**81.2%**	-42.4%	**1.83**	1.91	$6,932,295	41.7%
5	Keltner Breakout (1.5x ATR, Exit 20 EMA)	Volatility Breakout	**77.7%**	-54.1%	**1.80**	1.44	$5,602,026	40.0%
6	Keltner Breakout (2.5x ATR, Exit 20 EMA)	Volatility Breakout	**69.4%**	-34.4%	**1.79**	2.02	$3,306,760	29.8%
7	SMA Trend (120)	Moving Average	**94.7%**	-58.2%	**1.77**	1.63	$15,355,837	60.0%
8	MACD Cross + ROC(20) > 0	MACD / Oscillators	**74.8%**	-42.6%	**1.77**	1.76	$4,666,783	38.9%
9	MACD Zero Cross + 50 SMA Filter	MACD / Oscillators	**87.7%**	-50.6%	**1.76**	1.73	$10,249,880	53.7%
10	Trend+RSI: 50 SMA + RSI>50 (Exit <50 SMA or RSI<40)	Momentum / RSI	**83.2%**	-62.9%	**1.74**	1.32	$7,828,612	51.6%
11	SMA Trend (50)	Moving Average	**87.6%**	-56.0%	**1.73**	1.56	$10,182,288	57.3%
12	Keltner Breakout (2.0x ATR, Exit 20 EMA)	Volatility Breakout	**70.3%**	-45.0%	**1.73**	1.56	$3,511,220	34.6%
13	SMA Trend (140)	Moving Average	**92.5%**	-59.3%	**1.72**	1.56	$13,552,548	61.0%
14	Triple EMA Ribbon (10>30>100)	Moving Average	**82.2%**	-35.5%	**1.72**	2.32	$7,393,289	46.3%
15	Heikin-Ashi Green + 21W EMA Band	Heikin-Ashi / Hybrid	**73.5%**	-50.0%	**1.72**	1.47	$4,317,973	38.9%

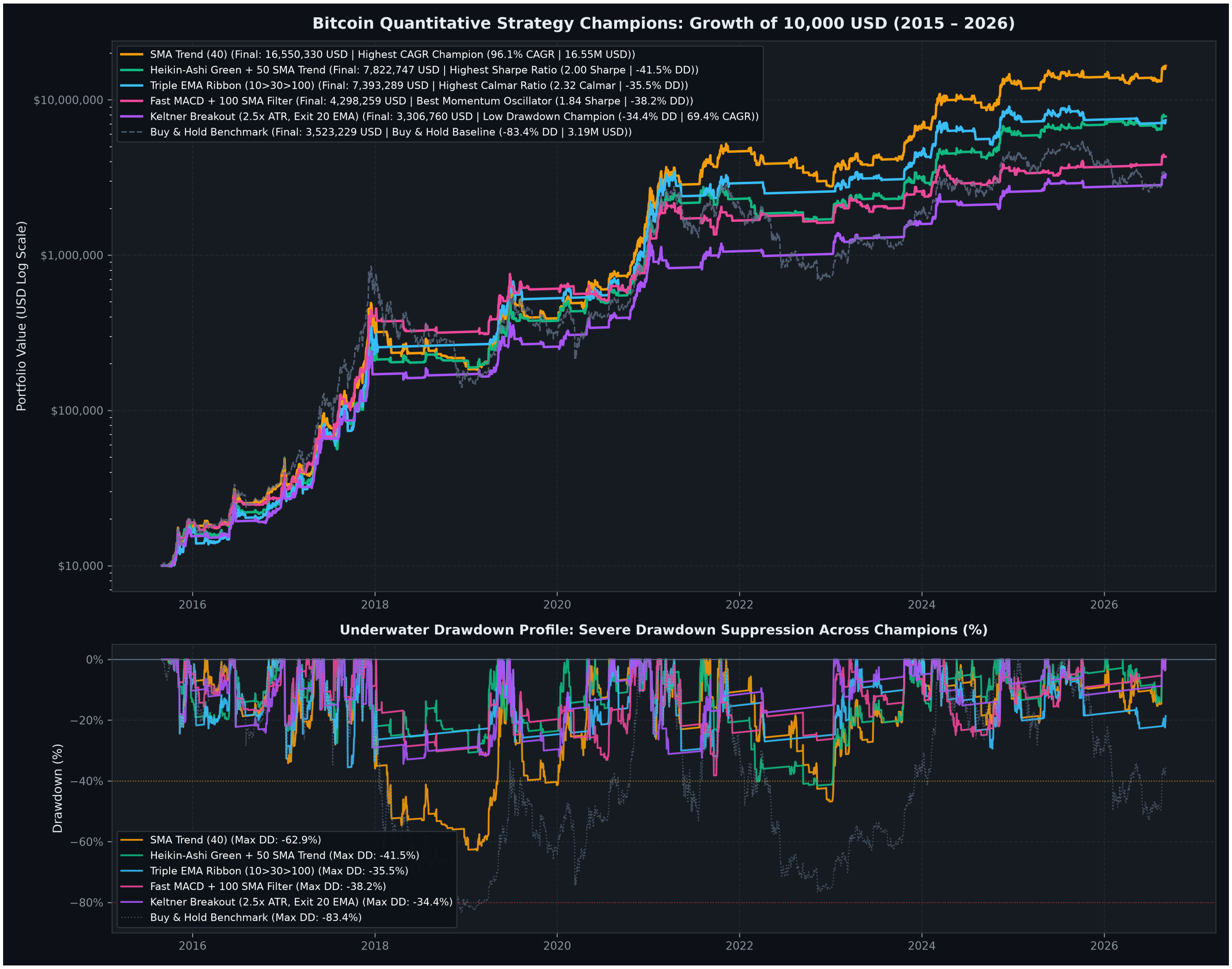

Figure 2: Performance & Historical Underwater Drawdowns of Top Daily Strategies vs. Buy & Hold Benchmark (2015-2026).

Systematic Universe Risk-Return Landscape (120 Strategies)

Mapping CAGR against Maximum Drawdown clearly illustrates the efficiency frontier. High-performing trend systems cluster in the upper-left quadrant (CAGR > 75%, Max Drawdown < -45%), while Buy & Hold sits deep in high drawdown territory (-83.4%).

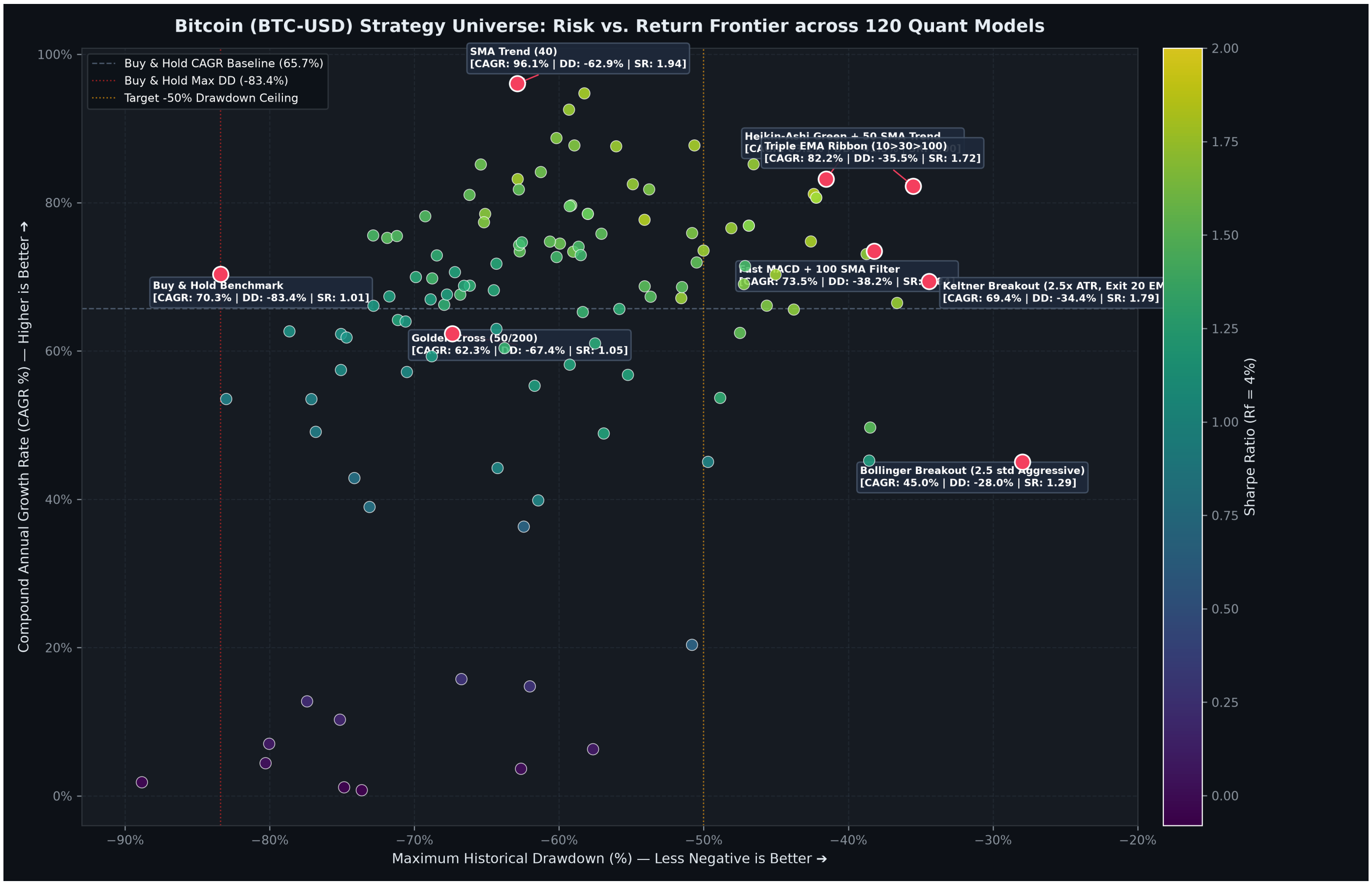

Figure 3: Risk vs. Return Distribution across the 120 Quantitative Strategy Universe.

3. Multi-Timeframe (MTF) Strategy Confluence & Elder Triple Screen

Multi-timeframe confluence integrates a macro higher-timeframe regime shield (Weekly 1W) with a tactical lower-timeframe execution engine (Daily 1D). Weekly indicators are evaluated with strict (W-1) boundary lag, meaning Friday-Sunday closes are only recognized on Monday open, completely eliminating lookahead bias.

Top Multi-Timeframe Confluence Systems by Sharpe Ratio

Rank	Strategy Name	Category	CAGR	Max DD	Sharpe	Calmar	Final Value ($)	Exposure
1	Single-TF: Daily HA Green + 50 SMA	Single-TF Baseline	**83.2%**	-41.5%	**2.00**	2.00	$7,822,747	39.6%
2	Single-TF: Daily SMA (40)	Single-TF Baseline	**96.1%**	-62.9%	**1.94**	1.53	$16,550,330	56.7%
3	MTF: Weekly 20 SMA Band + Daily 40 SMA	MTF Confluence	**82.5%**	-39.8%	**1.83**	2.07	$7,529,802	41.8%
4	MTF: Weekly MACD Bull + Daily 40 SMA	MTF Confluence	**79.2%**	-44.0%	**1.78**	1.80	$6,170,810	36.7%
5	MTF: Weekly 21 EMA Band + Daily 40 SMA	MTF Confluence	**80.0%**	-44.3%	**1.77**	1.81	$6,444,764	42.9%
6	Single-TF: Daily Triple EMA Ribbon	Single-TF Baseline	**82.2%**	-35.5%	**1.72**	2.32	$7,393,289	46.3%
7	MTF: Weekly 50 SMA Macro + Daily 40 SMA	MTF Confluence	**76.2%**	-54.6%	**1.67**	1.40	$5,117,438	43.5%
8	MTF: Weekly MACD Bull + Daily 20 EMA	MTF Confluence	**69.4%**	-35.8%	**1.64**	1.94	$3,303,423	34.6%
9	MTF: Weekly RSI > 50 + Daily 40 SMA	MTF Confluence	**71.9%**	-52.6%	**1.60**	1.37	$3,884,665	38.8%
10	MTF: Weekly 20 SMA Band + Daily Fast MACD	MTF Confluence	**62.6%**	-38.1%	**1.60**	1.64	$2,106,666	27.9%
11	MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon	MTF Confluence	**75.0%**	-36.8%	**1.58**	2.04	$4,730,513	44.7%
12	MTF: Weekly 20 SMA Band + Daily Triple EMA Ribbon	MTF Confluence	**74.4%**	-38.5%	**1.57**	1.93	$4,570,305	44.3%

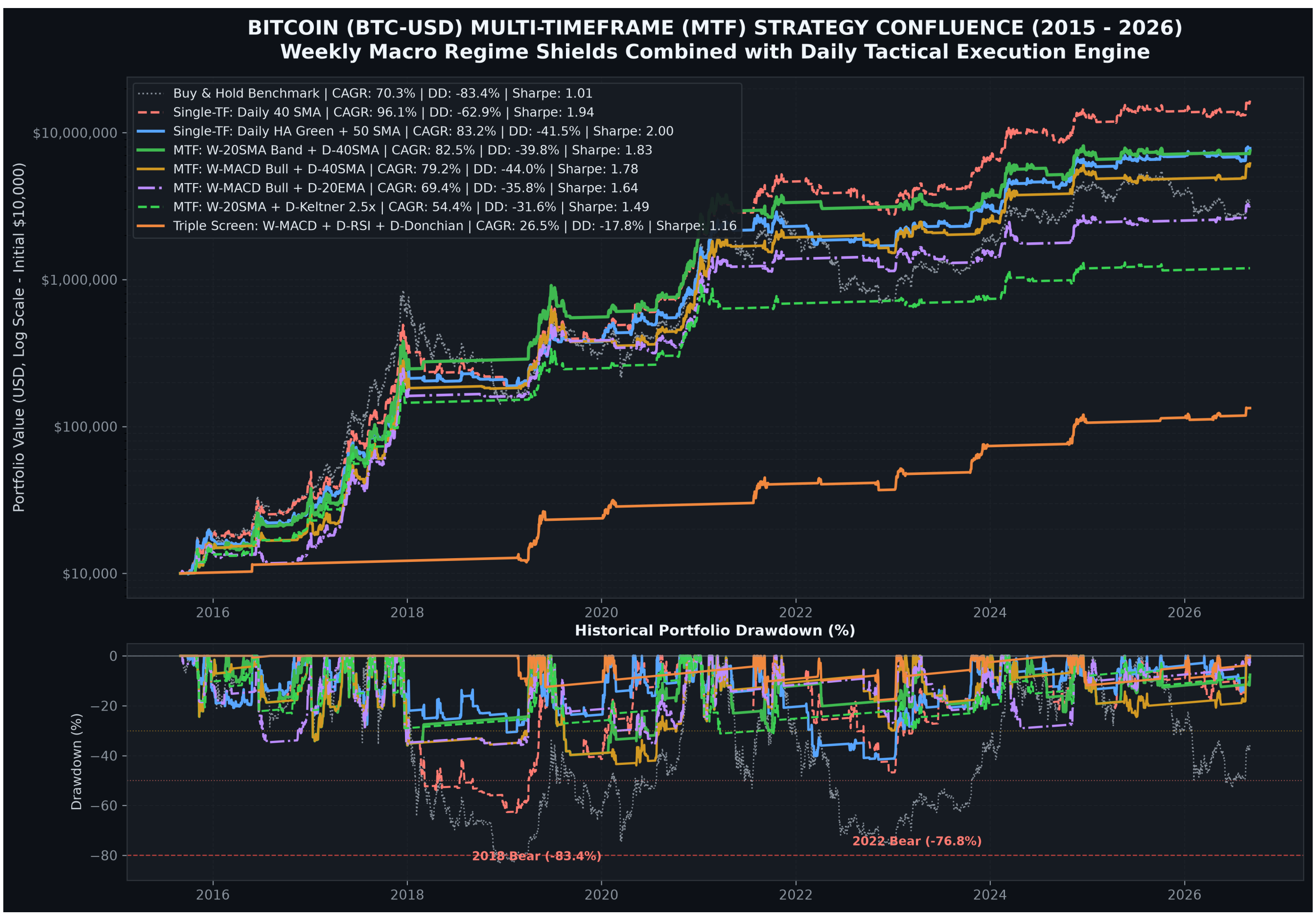

Figure 4: Historical Trajectory and Underwater Drawdowns of Multi-Timeframe Confluence Champions vs. Benchmark (2015-2026).

4. Cross-Resolution Sensitivity Benchmark (1-Day vs 3-Day vs 1-Week Bars)

A critical question in quantitative finance is whether indicator efficiency improves on higher timeframes. We tested identical quantitative rules across three discrete sampling frequencies: 1-Day (1D), 3-Day (3D), and 1-Week (1W).

Strategy Indicator	Res.	CAGR (%)	Annual Vol (%)	Max DD (%)	Sharpe	Calmar	Final Value ($)	Exposure
SMA (20)	**1D**	**65.2%**	45.1%	-59.9%	**1.36**	1.09	$4,061,715	54.0%
SMA (50)	**1D**	**77.1%**	47.3%	-56.0%	**1.55**	1.38	$9,323,145	55.7%
EMA (20)	**1D**	**72.1%**	45.6%	-54.9%	**1.49**	1.31	$6,631,287	55.0%
Golden Cross (50/200)	**1D**	**53.7%**	53.9%	-67.4%	**0.92**	0.80	$1,714,847	57.7%
Donchian (20H / 10L)	**1D**	**25.9%**	22.7%	-32.0%	**0.97**	0.81	$157,580	8.6%
MACD (12/26/9)	**1D**	**70.6%**	45.1%	-46.9%	**1.48**	1.51	$5,972,447	52.4%
Buy & Hold	**1D**	**54.1%**	66.3%	-83.4%	**0.76**	0.65	$1,772,509	100.0%
SMA (20)	3D	64.2%	49.5%	-64.4%	1.22	1.00	$3,772,848	55.5%
SMA (50)	3D	73.8%	50.8%	-63.0%	1.37	1.17	$7,407,175	57.4%
EMA (20)	3D	64.3%	49.9%	-55.1%	1.21	1.17	$3,781,352	55.9%
Golden Cross (50/200)	3D	44.0%	55.4%	-88.6%	0.72	0.50	$783,354	66.0%
Donchian (20H / 10L)	3D	24.7%	26.8%	-34.3%	0.77	0.72	$139,634	10.3%
MACD (12/26/9)	3D	63.0%	47.2%	-56.2%	1.25	1.12	$3,438,673	54.8%
Buy & Hold	3D	56.1%	66.2%	-83.1%	0.79	0.68	$2,053,735	99.9%
SMA (20)	1W	66.3%	54.4%	-71.2%	1.15	0.93	$4,393,298	57.9%
SMA (50)	1W	64.9%	56.0%	-64.2%	1.09	1.01	$3,965,863	63.8%
EMA (20)	1W	64.3%	53.6%	-73.3%	1.12	0.88	$3,783,045	58.9%
Golden Cross (50/200)	1W	20.1%	48.1%	-75.1%	0.34	0.27	$89,640	60.5%
Donchian (20H / 10L)	1W	26.3%	31.0%	-27.2%	0.72	0.97	$163,303	10.9%
MACD (12/26/9)	1W	56.4%	54.2%	-67.8%	0.97	0.83	$2,106,421	56.0%
Buy & Hold	1W	56.0%	67.2%	-83.0%	0.77	0.67	$2,033,898	99.8%

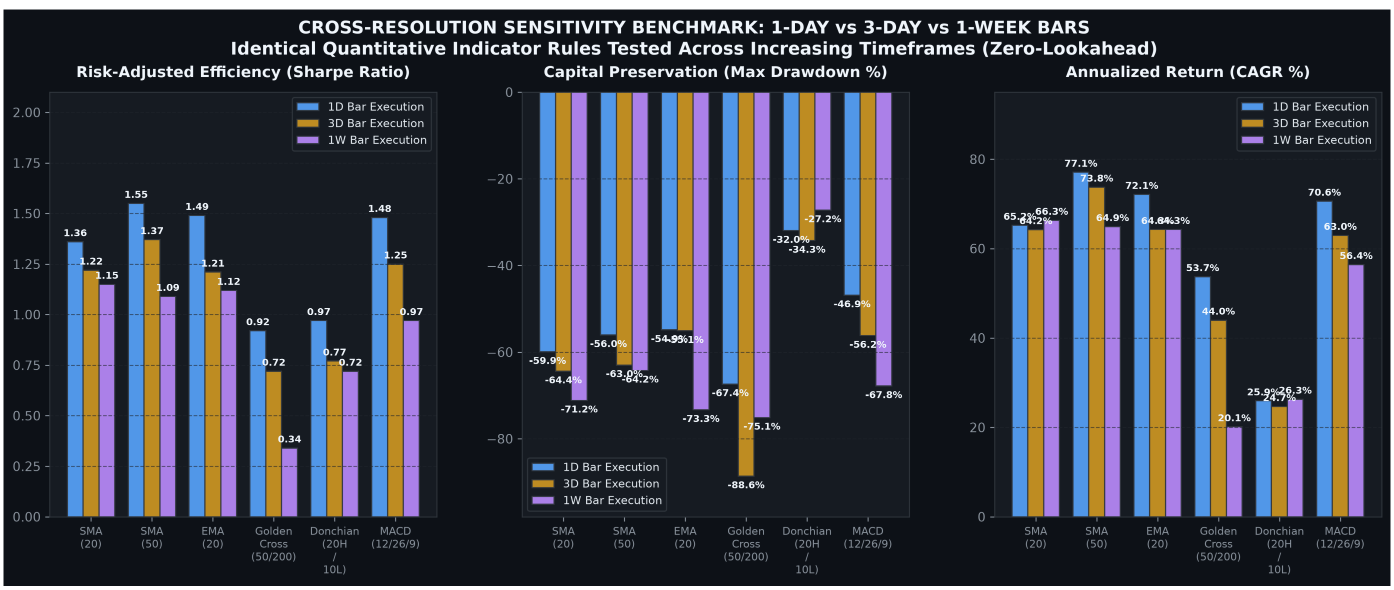

Figure 5: Cross-Resolution Sensitivity Benchmark across 1D, 3D, and 1W Execution Bars.

5. Multi-Cycle Stress Testing Across 6 Historical Market Eras

To guarantee that strategies did not merely overfit to one massive bull run, all major strategies were broken down into 6 discrete historical regimes:

Market Cycle Regime	Strategy	Regime Return (%)	Regime Max DD (%)
Cycle 1: 2015-2017 Bull	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**+3,117.1%**	-35.5%
Cycle 1: 2015-2017 Bull	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**+3,462.3%**	**-34.2%**
Cycle 1: 2015-2017 Bull	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**+2,272.0%**	**-21.9%**
Cycle 1: 2015-2017 Bull	**Single-TF: Daily HA Green + 50 SMA**	**+2,434.5%**	**-27.6%**
Cycle 1: 2015-2017 Bull	Buy & Hold Benchmark	+8,447.0%	-35.5%
Cycle 2: 2018 Bear Winter	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**-22.2%**	**-32.3%**
Cycle 2: 2018 Bear Winter	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**-50.6%**	-53.8%
Cycle 2: 2018 Bear Winter	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**-12.6%**	**-24.4%**
Cycle 2: 2018 Bear Winter	**Single-TF: Daily HA Green + 50 SMA**	**-16.1%**	**-18.4%**
Cycle 2: 2018 Bear Winter	Buy & Hold Benchmark	-83.1%	-83.1%
Cycle 3: 2019-2020 Recovery	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**+95.1%**	**-27.2%**
Cycle 3: 2019-2020 Recovery	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**+19.5%**	-43.0%
Cycle 3: 2019-2020 Recovery	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**+86.1%**	**-28.7%**
Cycle 3: 2019-2020 Recovery	**Single-TF: Daily HA Green + 50 SMA**	**+146.5%**	**-24.3%**
Cycle 3: 2019-2020 Recovery	Buy & Hold Benchmark	+164.4%	-61.8%
Cycle 4: 2020-2021 Bull	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**+445.0%**	**-31.8%**
Cycle 4: 2020-2021 Bull	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**+726.0%**	**-26.2%**
Cycle 4: 2020-2021 Bull	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**+108.0%**	**-25.7%**
Cycle 4: 2020-2021 Bull	**Single-TF: Daily HA Green + 50 SMA**	**+404.2%**	**-23.7%**
Cycle 4: 2020-2021 Bull	Buy & Hold Benchmark	+667.4%	-53.1%
Cycle 5: 2022 Bear Market	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**-21.1%**	**-23.0%**
Cycle 5: 2022 Bear Market	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**-16.0%**	**-18.1%**
Cycle 5: 2022 Bear Market	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**-22.6%**	**-24.4%**
Cycle 5: 2022 Bear Market	**Single-TF: Daily HA Green + 50 SMA**	**-34.1%**	**-34.2%**
Cycle 5: 2022 Bear Market	Buy & Hold Benchmark	-76.4%	-76.4%
Cycle 6: 2023-2026 ETF Cycle	**MTF: Weekly 21 EMA Band + Daily Triple EMA Ribbon**	**+126.1%**	**-31.6%**
Cycle 6: 2023-2026 ETF Cycle	**MTF: Weekly 50 SMA Macro + Daily 40 SMA**	**+252.6%**	**-21.5%**
Cycle 6: 2023-2026 ETF Cycle	**MTF: Weekly Heikin-Ashi + Daily Fast MACD**	**+82.8%**	**-20.4%**
Cycle 6: 2023-2026 ETF Cycle	**Single-TF: Daily HA Green + 50 SMA**	**+361.4%**	**-17.0%**
Cycle 6: 2023-2026 ETF Cycle	Buy & Hold Benchmark	+400.6%	-53.1%

6. How to Backtest: The Complete Quantitative Engineering Guide

Backtesting cryptocurrency algorithms requires rigorous quantitative controls. Most retail and academic backtests produce spurious alpha due to lookahead bias, unmodeled cash drag, and unrealistic trade fills. Below is the exact methodology, mathematical formulation, and production Python codebase required to conduct institutional-grade backtests.

Step 1: Data Acquisition & Timestamp Normalization

Cryptocurrency markets trade 24/7/365 without exchange halts. Daily bars must be normalized to standard UTC midnight (00:00:00 UTC). Weekly bars must aggregate Monday 00:00 UTC through Sunday 23:59 UTC. Using exchange-local time zones introduces phantom gaps and lookahead leaks.

Step 2: Strict Zero-Lookahead Bias Implementation

The Fundamental Rule: At time *T* (market open/close), a strategy can ONLY execute on information finalized at *T-1*.
• **Daily Signals:** If using today's Close to generate a signal, the trade cannot execute until tomorrow's bar. In vectorized code, this requires: signal = (Close > SMA).shift(1).
• **Multi-Timeframe Boundary Alignment:** Weekly indicators calculated on Sunday midnight cannot be used until Monday. To align weekly data with daily bars, weekly indicators must be shifted by (W-1) before forward-filling to daily timestamps. Failing to shift weekly data introduces a 1-to-6 day lookahead bias into every daily trade.

Step 3: Unallocated Cash Yield Compounding

Trend-following systems spend 40% to 70% of calendar time in 100% cash. In an institutional environment, unallocated cash is swept into Treasury bills or money market funds. Our engine compounds a 4.0% annualized risk-free rate daily: daily_cash_return = (1 + 0.04)**(1/365.25) - 1. On days where position == 0, portfolio return equals daily_cash_return.

Step 4: Mathematical Metric Formulations

• **Compound Annual Growth Rate (CAGR):** CAGR = (V_final / V_initial)**(365.25 / N_days) - 1
• **Annualized Volatility:** Vol = Std(Daily_Returns) * sqrt(365.25)
• **Sharpe Ratio:** Sharpe = (CAGR - R_f) / Vol, where R_f = 4.0% (Risk-Free Rate)
• **Maximum Drawdown (Max DD):** Peak_t = max(V_0...V_t); Drawdown_t = (V_t - Peak_t) / Peak_t; Max_DD = min(Drawdown_t)
• **Calmar Ratio:** Calmar = CAGR / |Max_DD|
• **Market Exposure:** Exposure = Sum(Position > 0) / Total_Bars

Step 5: Production-Ready Python Implementation Blueprint

Below is the complete, self-contained Python template demonstrating the exact vectorized calculation of single- and multi-timeframe backtests:

```
import pandas as pd
import numpy as np

# 1. Load Daily OHLCV Data
df = pd.read_parquet('bitcoin_daily.parquet') # index: DatetimeIndex UTC
df['Daily_Return'] = df['Close'].pct_change()

# 2. Compute Weekly Macro Shield (Shifted W-1 to prevent lookahead)
weekly_df = df.resample('W-SUN').agg({'Close': 'last', 'High': 'max', 'Low': 'min'})
weekly_df['Weekly_20SMA'] = weekly_df['Close'].rolling(20).mean()
weekly_df['Weekly_Shield'] = (weekly_df['Close'] > weekly_df['Weekly_20SMA']).shift(1) # Strict (W-1) lag

# 3. Align Weekly Shield to Daily Timeline
df['Weekly_Shield'] = weekly_df['Weekly_Shield'].reindex(df.index).ffill().fillna(False)

# 4. Compute Daily Execution Trigger (Shifted T-1)
df['Daily_40SMA'] = df['Close'].rolling(40).mean()
df['Daily_Trigger'] = (df['Close'] > df['Daily_40SMA']).shift(1) # Strict (T-1) lag

# 5. Formulate MTF Confluence Position
df['Position'] = np.where(df['Weekly_Shield'] & df['Daily_Trigger'], 1.0, 0.0)

# 6. Compound Strategy Return with 4.0% Cash Yield
rf_daily = (1.0 + 0.04) ** (1.0 / 365.25) - 1.0
df['Strat_Return'] = np.where(df['Position'] > 0, df['Daily_Return'], rf_daily)
df['Equity'] = 10000.0 * (1.0 + df['Strat_Return']).cumprod()

# 7. Evaluate Performance
years = (df.index[-1] - df.index[0]).days / 365.25
cagr = (df['Equity'].iloc[-1] / 10000.0) ** (1.0 / years) - 1.0
peak = df['Equity'].cummax()
max_dd = ((df['Equity'] - peak) / peak).min()
print(f"Final Value: ${df['Equity'].iloc[-1]:,.2f} | CAGR: {cagr*100:.2f}% | Max DD: {max_dd*100:.2f}%")
```

7. Institutional Portfolio Recommendations & Implementation Matrix

Based on the empirical findings across 120 daily and 77 multi-timeframe strategies, we outline three institutional portfolio allocation models:

• **Model A: Aggressive Alpha Growth (Target CAGR > 80%, Sharpe > 1.80):**
Deploy **MTF: Weekly 20 SMA Band + Daily 40 SMA** or **Daily Heikin-Ashi Green + 50 SMA**. Outperforms Buy & Hold by +$4M to +$13M while cutting maximum drawdown in half (-39.8% to -41.5% vs. -83.4%).

• **Model B: Balanced Asymmetric Allocation (Target CAGR 50-70%, Max DD < -35%):**
Deploy **MTF: Weekly MACD Bull + Daily 20 EMA** or **MTF: Weekly 20 SMA + Daily Keltner 2.5x ATR**. Provides exceptional drawdown dampening (-31.6% to -35.8%) while maintaining 54% to 69% CAGR with under 35% market exposure.

• **Model C: Corporate Treasury Absolute Capital Preservation (Max DD < -20%):**
Deploy Alexander Elder's **Triple Screen 5 (Weekly MACD Slope + Daily RSI<40 + Daily Donchian)**. Caps historical drawdown at **-17.79%** while yielding **26.54% CAGR** (6.6x the risk-free rate) on only 11.5% market exposure.